

Problem 1

(i) Rings : Yes. Identity : Yes $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

Commutative: No. $\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$

$$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = 0.$$

Zero divisor : Yes. $\xrightarrow{\quad}$

(ii) No. No identity for addition.

(iii) Ring: Yes. $(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$ Law of distribution.

Identity element : Yes. $X \cdot A = X \cap A = A$, $\phi + A = \phi \cup A = A$.

Commutative: Yes. $A \cap B = B \cap A$.

Zero divisor: Yes. Let $A \subsetneq X$. Then $A \cdot A^c = A \cap A^c = \phi$.

Problem 2.

(i) $x(-y) + xy = x(-y+y) = x \cdot 0 = 0 \Rightarrow x(-y) = -xy.$

Similarly, $(-x)y = -xy.$

(ii) $0 = 0 \cdot x = (1 + (-1)) \cdot x = 1 \cdot x + (-1) \cdot x = x + (-1) \cdot x \Rightarrow -x = (-1) \cdot x.$

(iii) $(x-y)z = (x+(-y)) \cdot z = xz + (-y)z = xz - yz.$
 $\uparrow$
 (i).

P3. (i) Yes.

(ii) No. $0 \cdot \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 0 & a \\ b & 0 \end{pmatrix} \neq \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$

(iii) No. $0 \cdot \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$

$$\parallel$$
$$(\lambda + (-\lambda)) \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} \lambda^{-1}a & b \\ c & \lambda^{-1}d \end{pmatrix} + \begin{pmatrix} -\lambda^{-1}a & b \\ c & -\lambda^{-1}d \end{pmatrix} = \begin{pmatrix} 0 & 2b \\ 2c & 0 \end{pmatrix}$$

P4. If $n = p \cdot q$ where $p, q > 1$, then $p, q < n$.

Hence, $\underbrace{1+1+\dots+1}_{p \text{ times}} \neq 0$, and $p \in F$ is invertible.

It follows that $0 = \underbrace{1+1+\dots+1}_{n \text{ times}} = \underbrace{p+p+\dots+p}_{q \text{ times}}.$

However, $p \in F$, $p^{-1} \in F$, showing that

$$0 = 0 \cdot p^{-1} = p^{-1}(p + \dots + p) = \underbrace{1+1+\dots+1}_{q \text{ times}},$$

Contradicting to the minimality of n .